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Intermediate Quantity Measurements

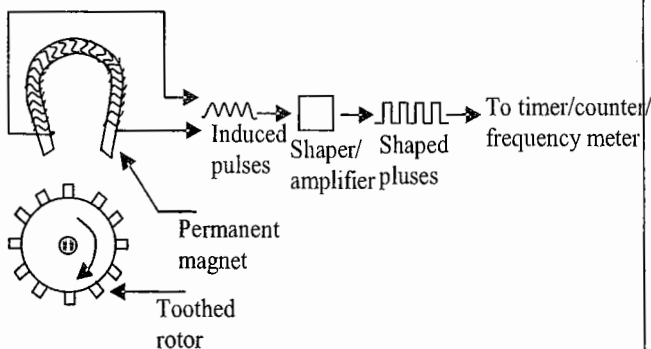
Measurements of Rotational Velocity

Contact less electrical tachometers:

- Tachometers of this type produce pulse from a rotating shaft without any physical contact between the speed transducer and the shaft. This aspect has the distinct advantage in that no load is applied to the machine.

Inductive pick-up tachometer

- The unit consists of a small permanent magnet with a coil round it. This magnetic pick up is placed near a metallic toothed rotor whose speed is to be measured. As the shaft rotates, the teeth pass in front of the pick-up and produce a change in the reluctance of the magnetic circuit. The field expands or collapses and a voltage is induced in the coil. The frequency of the pulses depends upon the number of teeth on the wheel and its speed of rotation. Since the number of teeth is known, the
- Speed of rotation can be determined by measuring the pulse frequency. To accomplish this task, pulse is amplified and squared, and fed into a counter of frequency measuring unit.



Inductive pick-up tachometer

- Let T be the number of teeth on the rotor, N be the revolutions per second and P be the number of pulses per second, then

$$\text{Speed } N = \frac{\text{pulses per second}}{\text{number of teeth}}$$

$$N = \frac{P}{T} \text{ rps} = \frac{P}{T} \times 60 \text{ rpm}$$

- If the rotor has 60 teeth, and if the counter counts the pulses in one second, then the counter will directly display the speed in revolutions per minute.

Example1:

A metallic toothed-rotor having 60 teeth was mounted on a shaft whose speed was to be determined. The pulses induced in the inductive-type magnetic pick-up were registered to be 3600 per second. Estimate the shaft speed. If the frequency meter gives the counts within $\pm 5\text{Hz}$, calculate the range within which the shaft speed can lie.

Ans: $(3600 \pm 5) \text{ rpm}$

Sol: Shaft speed is related to the number of teeth and the number of pulses by the relation.

$$\text{Speed} = \frac{\text{pulses per second}}{\text{number of teeth}}$$

$$= \frac{3600}{60} = 60 \text{ rps} = 3600 \text{ rpm}$$

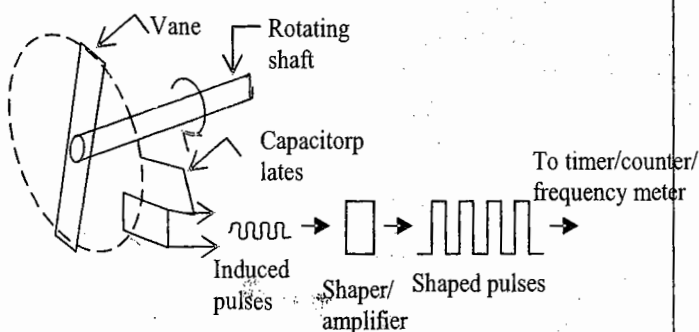
Accuracy of measurement = $\pm 5\text{Hz}$

$$= \pm \frac{5}{60} \times 60 = \pm 5 \text{ rpm}$$

$\therefore$ Range of shaft speed = $(3600 \pm 5) \text{ rpm}$

Capacitive type pick-up tachometer

The device consists of a vane attached to one end of the rotating **machine** shaft. When the shaft rotates between the fixed capacitive plates, there occurs a change in the capacitance. The capacitor forms a part of an oscillator tank so that number of frequency changes per unit of time is a measure of the shaft speed. The pulses thus produced are amplified, and squared, and may then be fed to frequency measuring unit or to a digital counter so as to provide a digital analog of the shaft rotation.



Capacitive pick-up tachometer

Measurement of Acceleration

There are two types of accelerometers generally used for measurement of acceleration:

- Piezo-electric type,
- Seismic type.

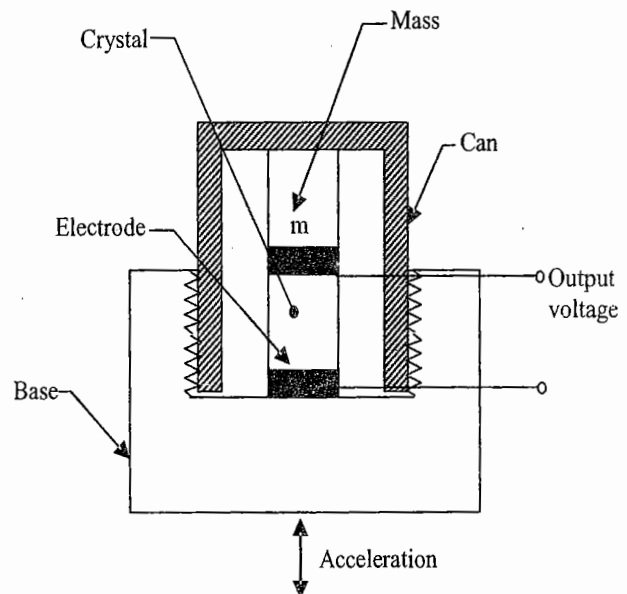
Piezo-electric accelerometer:

The unit is perhaps the simplest and most commonly used transducer employed for measuring acceleration. The sensor consists of a piezo-electric crystal sandwiched between two electrodes and has a mass placed on it. The unit is fastened to the base whose acceleration characteristics are to be obtained. The can threaded to the base acts as a spring and squeezes the mass against the crystal. Mass exerts a force

on the crystal and a certain output voltage is generated. If the base is now acceleration downward, inertial reaction force on the base acts upward against the top of the can. This relieves stress on the crystal. From Newton's second law

$$\text{Force} = \text{mass} \times \text{acceleration}$$

And since the mass is a fixed quantity, the decrease in force is in proportion to the acceleration. Likewise, an acceleration in the upward direction would increase the force on the crystal in proportion to the acceleration. The resulting change in the output voltage is recorded and correlated to the acceleration imposed on the base.



Piezo-electric accelerometer

Advantages and Limitations:

- Rugged and inexpensive device
- High output impedance
- High frequency response from 10 Hz to 50 kHz
- High sensitivity varies from 10 to 100 mV/g where $g = 9.807 \text{ m/s}^2$
- Capability to measure acceleration from a fraction of g to thousand of g



- Somewhat sensitive to changes in temperature
- Subject to hysteresis errors.

Displacement sensing (seismic) accelerometer:

In a seismic accelerometer, the displacement of a mass resulting from an applied force is measured and correlated to the acceleration. Fig. shows the schematics of a common spring-mass-damper system which accomplishes this task. The mass is supported by a spring and damper is connected to the housing frame. The frame is rigidly attached to the machine whose acceleration characteristics are to be determined. When an acceleration is imparted by the machine to the housing frame, the mass moves relative to the frame, and this relative displacement between the mass and frame is sensed and indicated by an electrical displacement transducer.

